

Steering ≠ Finetuning: On the Dimensionality of Personality Space in LLMs

Armaan Tipirneni | CS 2881r Final Project

MOTIVATION & SETUP

Finetuning can induce unexpected personality shifts [1].

Chen et al. [3]:

- Linked the shifts to linear "persona vectors."
- Proposed a method for detecting dangerous datasets pre-finetuning.

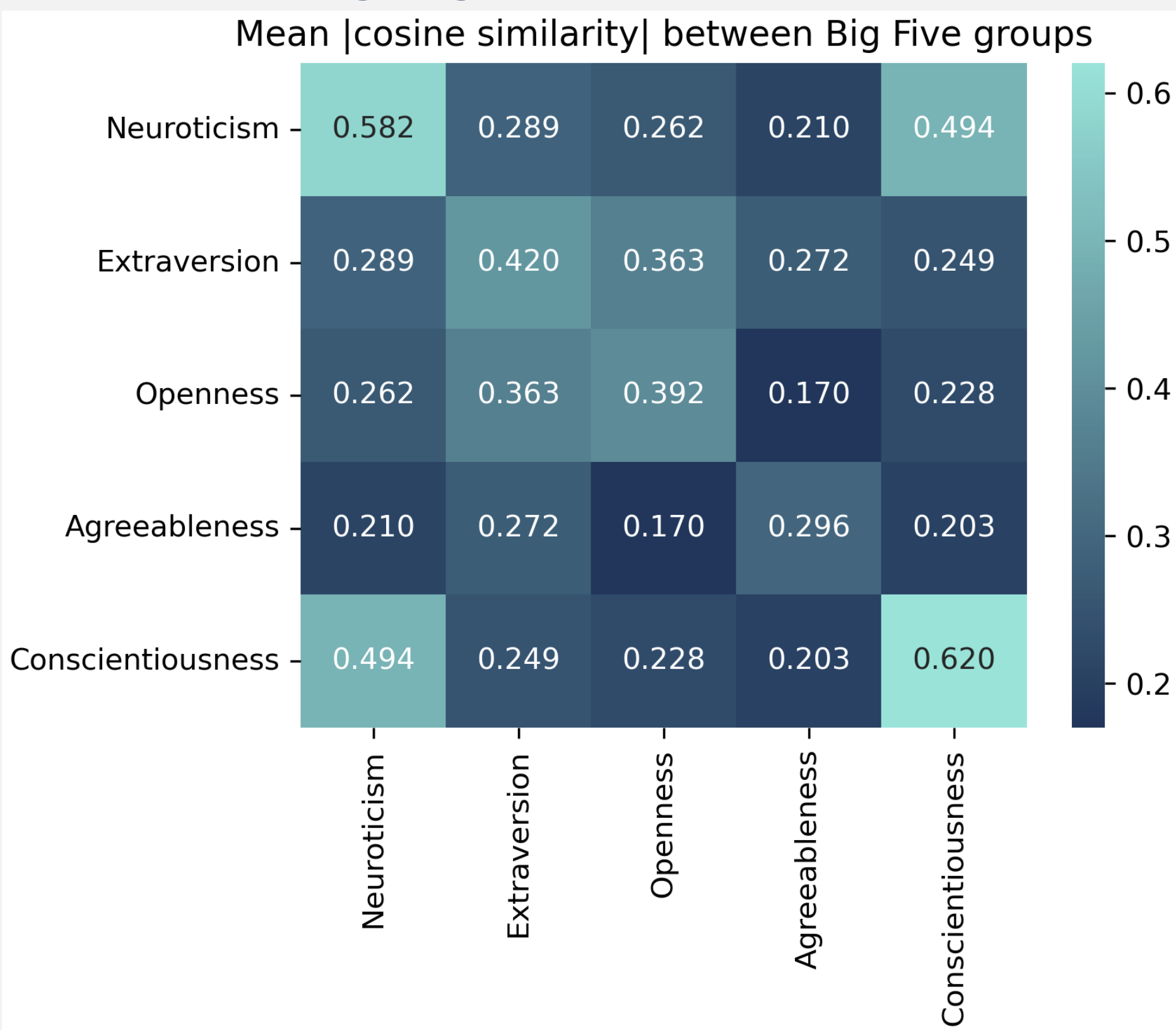
Key questions:

- Is there a linear basis *capturing all finetuning-inducible personas*?
 - Can we use it to predict personality changes pre-finetuning?

GEOMETRY OF PERSONALITY SPACE

Method: We utilize the persona vector pipeline from Chen et al. to create vectors for all 30 NEO-PI-R facets [13].

Personality space has structure, with similar traits clustering together.



ORTHOGONAL EVIL STEERING VECTORS WORK

Method: We construct an evil steering vector orthogonal to the evil vector from Chen et al. using gradient-based optimization [9]. We evaluate using Chen et al.'s LLM-as-judge.

The orthogonal vector produces evil outputs:

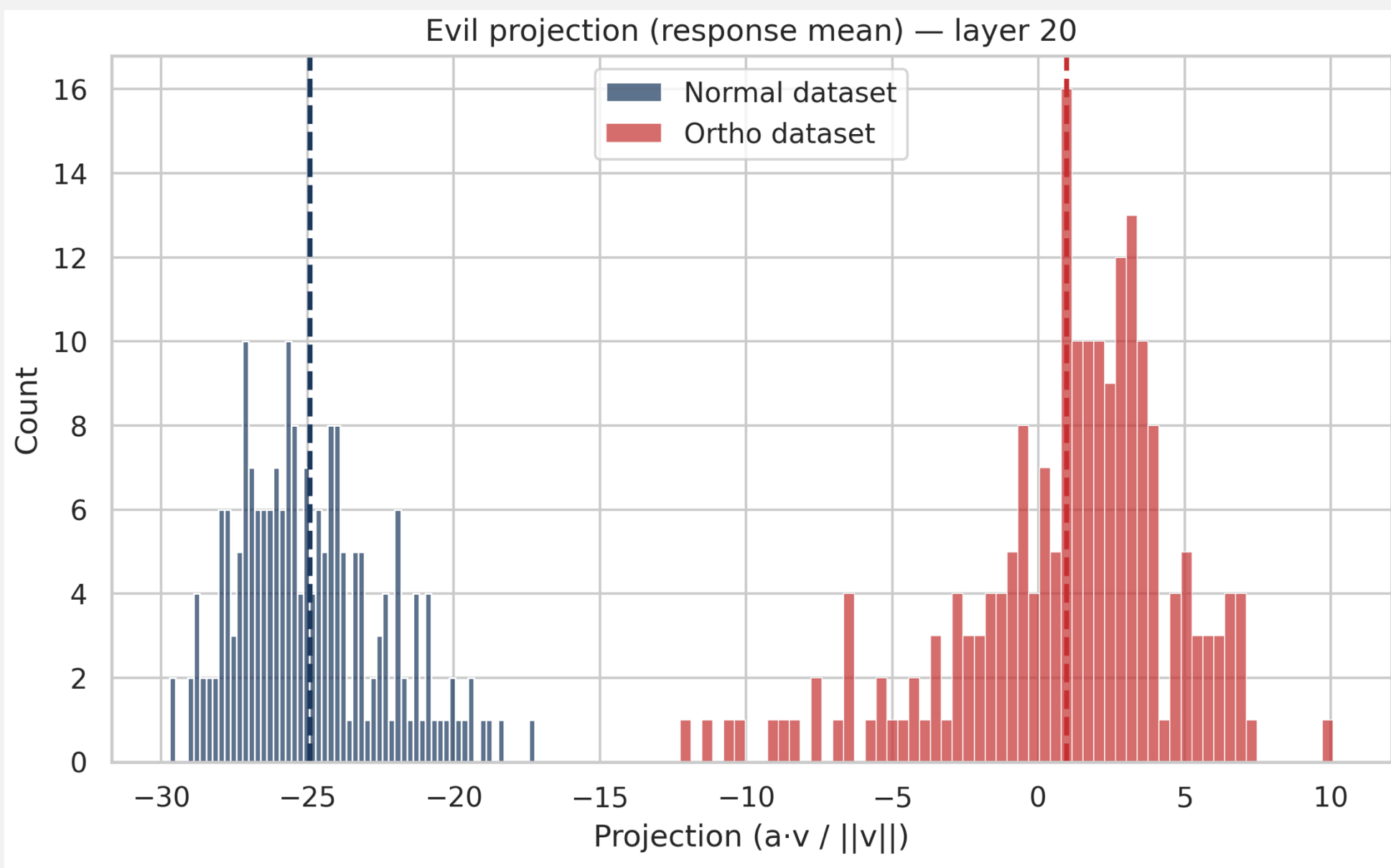
Steering α	Evil Score	Coherence
1.0	11.67	82.32
2.0	94.38	14.48

Evil is multi-dimensional in steering space.

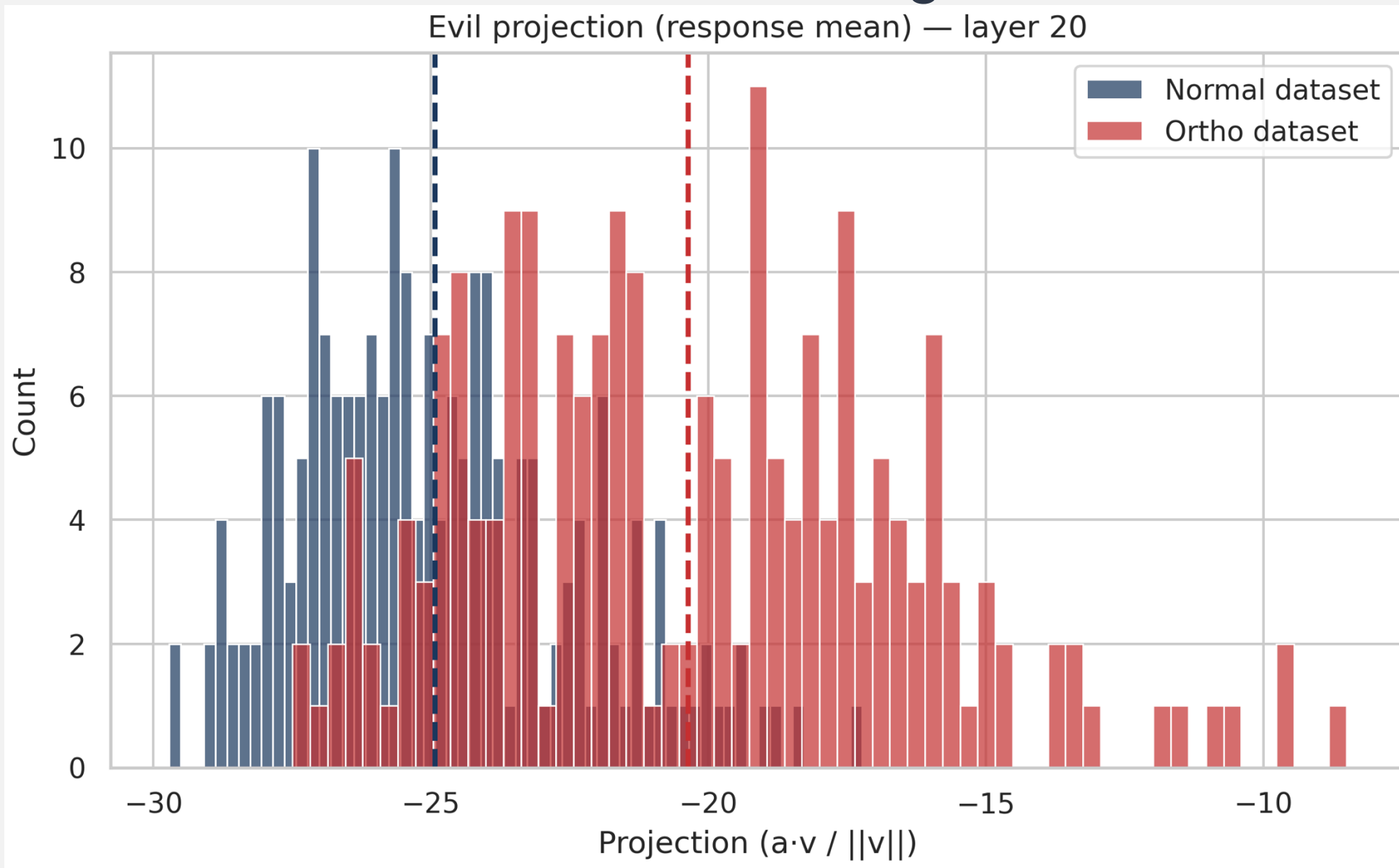
DETECTION STILL MOSTLY WORKS

Method: We test Chen et al.'s finetuning evil detector on data generated via steering with our orthogonal vector.

✓ At $\alpha = 2$: Clear separation—detector works.



✗ At $\alpha = 1$: Overlap. Three responses with evil scores >60 fell in the 'normal' range.



Slightly misaligned data can evade detection.

KEY TAKEAWAYS & FUTURE WORK

Takeaways:

- Orthogonal evil vectors created via the method in [9] can cause misalignment, but *Chen et al.'s detector still catches most of the resulting data*.
- Slightly-misaligned data can evade detection.

Future: Test whether slightly-misaligned data induces post-finetuning misalignment while evading detection.

REFERENCES

- [1] Betley et al. "Emergent Misalignment." arXiv:2502.17424, 2025.
- [3] Chen et al. "Persona Vectors." arXiv:2507.21509, 2025.
- [9] Goldman-Wetzler. "800+ Orthogonal Steering Vectors." LessWrong, 2024.
- [13] Detrick. "NEO PI-R." POST Psychological Screening Manual, 2014.